MMDU MULLANA OFFICIAL SYLLABUS 2025-26

Mathematics-I

Complete 4-Unit Lecture Notes & University Solved Examination Bank

Course Code: BMAT-001

Units Covered: Units 1, 2, 3, and 4 (Exhaustive Theory + Solved Problems)

Mathematical Rigor: Publication-grade KaTeX typesetting

Maharishi Markandeshwar (Deemed to be University), Mullana • 100% Offline Accessible

UNIT 1 • MMDU MULLANA ACADEMIC CURRICULUM

Unit 1: Unit I: Matrices (10 Hours)

Exam Weightage: 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Estimated Study Time:** 35 mins

Complete Unit I coverage per MMDU BMAT-001 syllabus: Rank of a matrix, Gauss-Jordan's Method for inverse of matrix, Normal form of Matrix; Consistency of system of linear equations and their solution; Eigenvalues and eigenvectors; Cayley-Hamilton Theorem and applications.

1. Rank of a Matrix & Normal (Canonical) Form

1. In-Depth Mathematical Foundations of Matrix Rank

✅ **FUNDAMENTAL LAW / RULE:**

Rigorous Linear Algebra Definition: Let A be an $m \times n$ matrix with entries in $\mathbb{R}$ (or $\mathbb{C}$). The **Rank** of A, denoted by $\rho(A)$ or $\text{rank}(A)$, is defined as the dimension of the vector space spanned by its row vectors (Row Rank) or column vectors (Column Rank). By the Fundamental Rank Theorem of Linear Algebra:

$$\text{Row Rank}(\mathbf{A}) = \text{Column Rank}(\mathbf{A}) = \rho(\mathbf{A}) \leq \min(\mathbf{m}, \mathbf{n})$$

Minor-Based Criterion of Rank:

A matrix A is said to be of rank r if and only if:

1. There exists **at least one minor** (determinant of a square submatrix) of order r that is strictly non-zero: $|M_r| \neq 0$.

2. **Every minor** of order $(r+1)$ and higher vanishes identically: $|M_{r+1}| = 0$.

---

2. Deep-Dive: Row Echelon Form Reduction

Strict Mathematical Criteria for Row Echelon Form:

A matrix E is in **Row Echelon Form** if:

1. **Zero Rows at Bottom:** All rows consisting entirely of zeros are grouped at the very bottom.

2. **Leading Entry (Pivot):** The leading coefficient (the first non-zero entry from the left) of each non-zero row is strictly to the right of the leading coefficient of the row directly above it.

3. **Zeros Below Pivots:** All entries in a column directly below a pivot entry are zero.

> $$\rho(\mathbf{A}) = \text{Total Number of Non-Zero Rows in Row Echelon Form}$$

Invariance of Rank under Elementary Row Transformations:

Elementary row transformations do not alter the linear span of the row vectors:

1. Interchanging two rows: $R_i \leftrightarrow R_j$ (Permutation matrix P, $|P| = -1 \neq 0$).

2. Multiplying a row by a non-zero scalar: $R_i \rightarrow kR_i,\ k \neq 0$ (Diagonal scaling matrix D, $|D| = k \neq 0$).

3. Adding a scalar multiple of one row to another: $R_i \rightarrow R_i + cR_j$ (Shear matrix S, $|S| = 1 \neq 0$).

Because all elementary matrices are non-singular ($|E| \neq 0$), rank is strictly invariant: $\rho(EA) = \rho(A)$.

---

3. Deep-Dive: Normal (Canonical) Form $[I_r\ 0; 0\ 0]$

By using **both elementary row AND elementary column operations**, any arbitrary $m \times n$ matrix A of rank r can be brought into its unique Canonical Normal Form:

> $$\mathbf{N} = \begin{bmatrix} I_r & 0_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{bmatrix}$$

Existence of Non-Singular Matrices P and Q:

There exist non-singular square matrices P (of order m) and Q (of order n) representing the product of all row operations and column operations respectively, such that:

$$\mathbf{P \cdot A \cdot Q} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

- **Procedure to compute P and Q simultaneously:**

1. Write $A_{m\times n} = I_m \cdot A \cdot I_n$.
2. Perform elementary row operations on A and simultaneously on the pre-factor I_m.
3. Perform elementary column operations on A and simultaneously on the post-factor I_n.
4. When the central matrix reduces to $[I_r\ 0; 0\ 0]$, the left matrix becomes P and the right matrix becomes Q!

---

4. Fundamental Properties & Sylvester's Rank Inequalities

1. **Sub-multiplicative Property:** $\rho(AB) \leq \min(\rho(A), \rho(B))$.

2. **Sylvester's Law of Nullity:** For an $m \times n$ matrix A and $n \times p$ matrix B:

$$\rho(\mathbf{A}) + \rho(\mathbf{B}) - \mathbf{n} \leq \rho(\mathbf{AB}) \leq \min(\rho(\mathbf{A}), \rho(\mathbf{B}))$$

3. **Rank of Transpose & Conjugate:** $\rho(A^T) = \rho(A)$ and $\rho(A^T A) = \rho(AA^T) = \rho(A)$.

4. **Rank of Sum:** $\rho(A + B) \leq \rho(A) + \rho(B)$.

Problem:
Find the rank of matrix $A = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}$.

Step 1 🚀: Compute the Determinant of A

$$|A| = (2)(6) - (4)(3) = 12 - 12 = 0$$

Since $|A| = 0$, the rank cannot be 2: $\rho(A) < 2$.

Step 2 ⚡: Check Order-1 Minors (Single Entries)
The matrix contains non-zero entries (e.g., $a_{11} = 2 \neq 0$). Therefore, there is at least one non-vanishing minor of order 1.

Step 3 🎯: Row Echelon Verification ($R_2 \to R_2 - 1.5R_1$)

$$\begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 \\ 0 & 0 \end{bmatrix}$$

There is exactly **1 non-zero row**.

🎯 Rank of A: $\rho(\mathbf{A}) = \mathbf{1}$.

University Question (4 Marks):
Reduce the matrix A to Row Echelon form and determine its rank:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$$

Step 1 🚀: Eliminate entries below pivot in Column 1
Apply row operations $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2 ⚡: Eliminate entry below pivot in Column 3
Apply $R_3 \to R_3 - R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 3 🎯: Count Non-Zero Rows
The echelon form contains 2 non-zero rows ($[1, 2, 3]$ and $[0, 0, 1]$) and 1 zero row.

🎯 Row Echelon Rank: $\rho(\mathbf{A}) = \mathbf{2}$.

University Question (6 Marks - MMDU Annual Exam):

Reduce matrix A to its Normal Form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ using elementary row AND column transformations, and find non-singular matrices P and Q such that $PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

Step 1 🚀: Setup Identity Matrices $A = I_3 A I_3$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2 ⚡: Elementary Row Operation $R_2 \to R_2 - R_1$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A I_3$$

Step 3 🎯: Elementary Column Operations $C_2 \to C_2 - C_1,\ C_3 \to C_3 - 2C_1$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 4 🏁: Row Operation $R_3 \to R_3 - R_2$ **and Column Operation** $C_3 \to C_3 - C_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

This is the standard Normal Form $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$.

🎯 Canonical Normal Form: $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$ | Rank $\rho(\mathbf{A}) = \mathbf{2}$ | $P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$, $Q = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$.

2. Gauss-Jordan Method for Matrix Inverse [A | I] ~ [I | A^{-1}]

1. The Gauss-Jordan Inversion Algorithm (Theoretical Analysis)

> 📌 **CRITICAL EXAM DERIVATION:**
> **University Exam Distinction:** Never use the classical Adjugate formula $A^{-1} = \frac{\text{adj}(A)}{|A|}$ when the question explicitly states **"Find inverse using Gauss-Jordan method"**! The classical method requires $O(n!)$ operations via minors, whereas Gauss-Jordan requires only $O(n^3)$ row operations.

Mathematical Principle of Gauss-Jordan Inversion:

Let A be an $n \times n$ invertible matrix ($|A| \neq 0$). If a sequence of elementary row operations $E_1, E_2, \dots, E_k$ transforms A into the identity matrix I:

$$(E_k \dots E_2 E_1)A = I \implies \mathbf{A^{-1} = E_k \dots E_2 E_1}$$

Therefore, if we apply the identical sequence of operations to the augmented matrix $[A \mid I_n]$:

$$(E_k \dots E_2 E_1)[A \mid I_n] = [(E_k \dots E_1)A \mid (E_k \dots E_1)I_n] = [\mathbf{I_n \mid A^{-1}}]$$

---

2. Step-by-Step Algorithmic Execution Pipeline

1. **Augmentation:** Construct the partitioned block matrix $[A \mid I_n]$ of dimension $n \times 2n$.

2. **Forward Elimination (Upper Triangularization):**

- For column $j = 1, 2, \dots, n$:
- **Pivoting:** Find row $i \geq j$ with a non-zero pivot a_{ij} (preferring 1 or -1). Swap $R_j \leftrightarrow R_i$.
- If pivot entry $a_{jj} = 0$, matrix is **singular** (A^{-1} does not exist!).
- Normalize the pivot row: $R_j \to \frac{1}{a_{jj}} R_j$.
- Eliminate all entries below pivot: for each $k > j$, $R_k \to R_k - a_{kj} R_j$.

3. **Backward Elimination (Diagonalization):**

- Starting from the bottom row $j = n, n-1, \dots, 2$:
- For each row $k < j$, eliminate the above-pivot entries: $R_k \to R_k - a_{kj} R_j$.

4. **Extraction:** The right half of the augmented matrix is strictly A^{-1}. Verify that $A \cdot A^{-1} = I_n$.

---

3. Comparison of Matrix Inversion Techniques

Feature	Gauss-Jordan Method	Adjugate Formula	LU Decomposition
Operation Count	n^3 additions & multiplications	$\sim (n+1)!$ determinant operations	$\frac{2}{3}n^3$ operations
Memory Required	$n \times 2n$ augmented grid	High cofactor matrix cache	$n \times n$ in-place factorization
Suitability	Hand calculation in university exams	2×2 matrices only	High-performance computer algorithms

---

Problem:
Find the inverse of matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ using the Gauss-Jordan method.

Step 1 🚀: Setup Augmented Matrix $[A \mid I]$

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right]$$

Step 2 ⚡: Make entry below pivot zero ($R_2 \to R_2 - 3R_1$)

$$\sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right]$$

Step 3 🎯: Normalize pivot of row 2 ($R_2 \to -\frac{1}{2}R_2$)

$$\sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

Step 4 🏁: Eliminate entry above pivot ($R_1 \to R_1 - 2R_2$)

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

🎯 Final Inverse: $\mathbf{A^{-1}} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$. Verified: $A \cdot A^{-1} = I$.

University Question (6 Marks - MMDU Question Paper):
Employ the Gauss-Jordan method to determine the inverse of:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Step 1 🚀: Swap Rows to Get Leading Pivot 1 ($R_1 \leftrightarrow R_2$)

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Step 2 ⚡: Eliminate Column 1 ($R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & -3 & -1 & 1 & -2 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \end{array}\right]$$

Step 3 🎯: Swap $R_2 \leftrightarrow R_3$, Normalize and Eliminate

$$\begin{aligned} R_2 \to -R_3: &\quad [0,1,-1 \mid 0,1,-1] \\ R_3 \to R_3 + 3R_2: &\quad [0,0,-4 \mid 1,1,-3] \\ R_3 \to -\frac{1}{4}R_3: &\quad [0,0,1 \mid -\frac{1}{4},-\frac{1}{4},\frac{3}{4}] \end{aligned}$$

Step 4 🏁: Eliminate Above Pivots to reach $[I \mid A^{-1}]$

$$\begin{aligned} R_2 \to R_2 + R_3: &\quad [0,1,0 \mid -\frac{1}{4},\frac{3}{4},-\frac{1}{4}] \\ R_1 \to R_1 - R_3 - 2R_2: &\quad [1,0,0 \mid \frac{3}{4},-\frac{1}{4},-\frac{1}{4}] \end{aligned}$$

🎯 Complete Inverse: $\mathbf{A^{-1}} = \frac{1}{4}\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$. Verified: $|A| = 4 \neq 0$.

3. Consistency of System of Linear Equations AX = B

1. Comprehensive Theory: System of Linear Equations $AX = B$

A system of m linear equations in n unknowns $x_1, x_2, \ldots, x_n$ is represented in matrix notation as:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \iff \mathbf{AX = B}$$

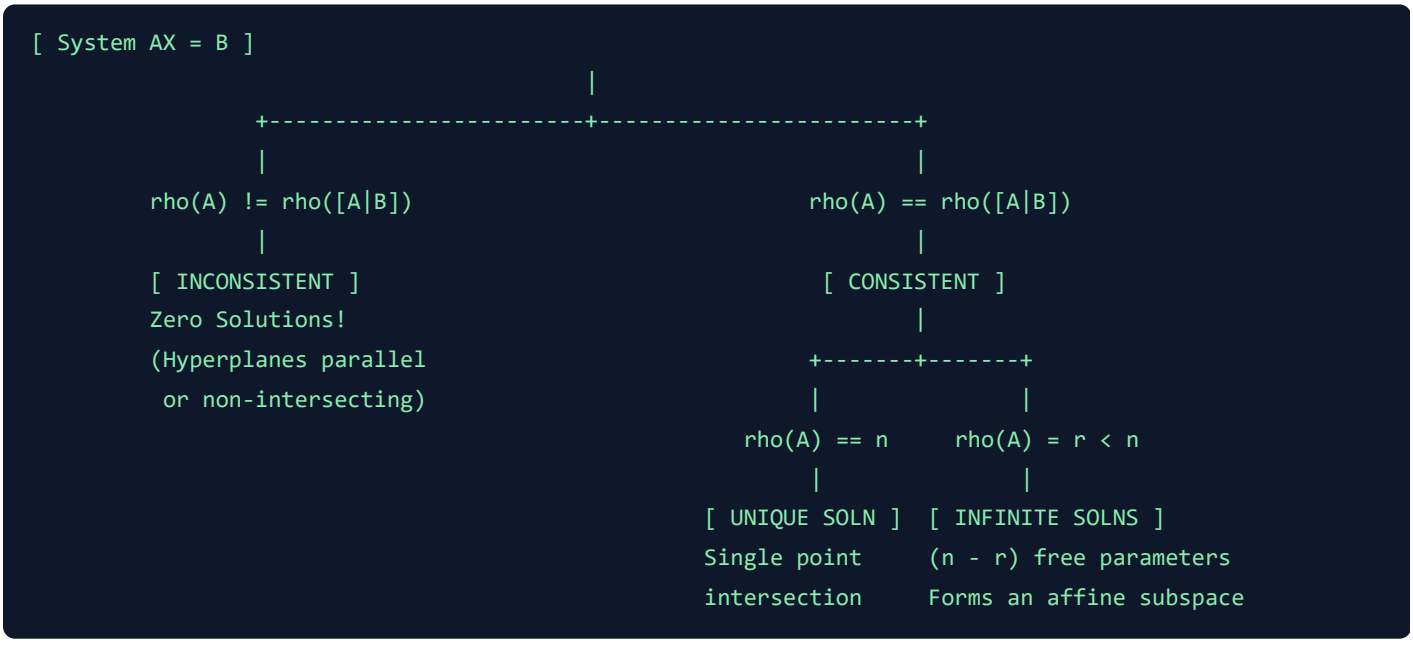
The Rouché–Capelli Theorem:

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: A system of linear equations $AX = B$ is **consistent** (possesses at least one solution) if and only if the rank of the coefficient matrix A equals the rank of the augmented matrix $[A \mid B]$:

$$\rho(\mathbf{A}) = \rho([\mathbf{A} \mid \mathbf{B}])$$

---

2. Complete Classification of Solution Regimes



---

3. Homogeneous Linear Systems $AX = 0$

For $AX = 0$, $B = 0$. Since augmenting a column of zeros cannot increase matrix rank, $\rho(A) = \rho([A \mid 0])$ always holds. Therefore, **a homogeneous system is ALWAYS consistent** and always possesses at least the trivial solution $X = 0$.

Nature of Solutions for Square Systems ($n \times n$):

1. **Trivial Solution ($X = 0$):**

- Occurs when $\rho(A) = n \iff |A| \neq 0$ (Non-singular matrix).
- The columns of A are linearly independent; the only solution is $x_1 = x_2 = \cdots = x_n = 0$.

2. **Non-Trivial Solutions (Infinite Non-Zero Solutions):**

- Occurs when $\rho(A) = r < n \iff |A| = 0$ (Singular matrix), yielding $(n - r)$ linearly independent parameters.

> 🚨 **EXAMINER TRAP / PITFALL:**
> **Examiner's Trap:** When an exam problem asks: "*Find values of k for which the homogeneous equations have non-trivial (non-zero) solutions*", simply set the determinant to zero: $\mathbf{|A| = 0}$!

University Question (6 Marks - Classic MMDU Exam Problem):
Investigate for what values of λ and μ the system of equations:

$$\begin{cases} x + y + z = 6 \\ x + 2y + 3z = 10 \\ x + 2y + \lambda z = \mu \end{cases}$$

has: (i) No solution, (ii) A unique solution, (iii) Infinitely many solutions.

Step 1 🚀: Setup Augmented Matrix and Echelon Reduction

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{array}\right]$$

Step 2 ⚡: Case Analysis by Rouché–Capelli Theorem

1. Case (i) No Solution (Inconsistent): $\lambda = 3,\ \mu \neq 10 \implies \rho(A) = 2 \neq \rho([A \mid B]) = 3$
2. Case (ii) Unique Solution: $\lambda \neq 3,\ \forall \mu \implies \rho(A) = \rho([A \mid B]) = 3 = n$
3. Case (iii) Infinite Solutions: $\lambda = 3,\ \mu = 10 \implies \rho(A) = \rho([A \mid B]) = 2 < n\ (z = k)$

🎯 **Verified Result:** (i) No solution for $\lambda = 3, \mu \neq 10$ | (ii) Unique for $\lambda \neq 3$ | (iii) Infinite for $\lambda = 3, \mu = 10$.

4. Eigenvalues, Eigenvectors & Cayley-Hamilton Theorem

1. Eigenvalues (Characteristic Roots) & Eigenvectors

✅ **FUNDAMENTAL LAW / RULE:**
Definition: Let A be an $n \times n$ square matrix. A scalar λ is called an **Eigenvalue** (or characteristic root / latent root) of A if there exists a **non-zero** column vector $X \neq 0$ such that:

$$\mathbf{AX = \lambda X \iff (A - \lambda I)X = 0}$$

The non-zero vector X is called the **Eigenvector** (or characteristic vector) corresponding to λ.

Characteristic Equation & Matrix Invariants (3×3 Shortcut):

For $AX = \lambda X$ to have a non-trivial solution $X \neq 0$, the matrix $(A - \lambda I)$ must be singular:

$$\mathbf{|A - \lambda I| = 0}$$

For a 3×3 matrix A, the characteristic polynomial expands directly using trace and principal minors:

$$\mathbf{\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0}$$

Where:

- $S_1 = \text{Tr}(A) = a_{11} + a_{22} + a_{33} = \lambda_1 + \lambda_2 + \lambda_3$ *(Sum of main diagonal elements = Sum of eigenvalues)*
- $S_2 = M_{11} + M_{22} + M_{33}$ *(Sum of cofactors/minors of main diagonal elements)*
- $|A| = \det(A) = \lambda_1 \cdot \lambda_2 \cdot \lambda_3$ *(Determinant = Product of eigenvalues)*

Key Properties of Eigenvalues:

1. The eigenvalues of an **upper or lower triangular (or diagonal) matrix** are simply its main diagonal elements.

2. The eigenvalues of a **real symmetric matrix** ($A^T = A$) are always **purely real**, and eigenvectors of distinct eigenvalues are mutually **orthogonal** ($X_i^T X_j = 0$).

3. If λ is an eigenvalue of A, then:

- λ^k is an eigenvalue of A^k.
- $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} (for $|A| \neq 0$).
- $\frac{|A|}{\lambda}$ is an eigenvalue of $\text{adj}(A)$.

---

2. Cayley-Hamilton Theorem & Its Applications

> ✅ **FUNDAMENTAL LAW / RULE:**
> **Theorem Statement: "Every square matrix satisfies its own characteristic equation."**

> If the characteristic equation of an $n \times n$ matrix A is:

> $$|A - \lambda I| = (-1)^n (\lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} + \cdots + c_n) = 0$$

> Then replacing scalar λ by matrix A and constant c_n by $c_n I$ yields the zero matrix:

> $$\mathbf{A^n + c_1 A^{n-1} + c_2 A^{n-2} + \cdots + c_n I = 0}$$

Two Primary Applications of Cayley-Hamilton Theorem:

1. **Finding Matrix Inverse A^{-1} (when $c_n = (-1)^n |A| \neq 0$):**

Multiply the matrix equation by A^{-1} and isolate A^{-1}:

$$\mathbf{A^{-1} = -\frac{1}{c_n}\left(A^{n-1} + c_1 A^{n-2} + \cdots + c_{n-1} I\right)}$$

2. Computing Higher Matrix Powers ($A^4 A^8$, etc.):

Express A^n in terms of lower powers $(A^{n-1}, \ldots, A, I)$ and substitute iteratively.

University Question (6 Marks - MMDU Annual Exam):
For matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$: (1) Verify Cayley-Hamilton Theorem, (2) Evaluate A^{-1}, and (3) Compute A^4.

Step 1 🚀: Characteristic Equation via Invariants $S_1, S_2, |A|$

- $S_1 = \text{Tr}(A) = 1 + 1 + 1 = 3$
- $S_2 = M_{11} + M_{22} + M_{33} = (1-1) + (1-3) + (1-0) = 0 - 2 + 1 = -1$
- $|A| = 1(0) - 2(3) + 1(-3) = -9$
- Characteristic equation: $\lambda^3 - 3\lambda^2 - \lambda + 9 = 0$.

Step 2 ⚡: Verify $A^3 - 3A^2 - A + 9I = 0$

$$A^2 = \begin{bmatrix} 4 & 3 & 0 \\ -3 & 2 & -2 \\ 6 & 4 & 5 \end{bmatrix}, \quad A^3 = A^2 \cdot A = \begin{bmatrix} 4 & 11 & 1 \\ -9 & -2 & -7 \\ 21 & 11 & 7 \end{bmatrix}$$

Substituting into $A^3 - 3A^2 - A + 9I$:

$$\begin{bmatrix} 4-12-1+9 & 11-9-2+0 & 1-0-1+0 \\ -9+9-0+0 & -2-6-1+9 & -7+6+1+0 \\ 21-18-3+0 & 11-12+1+0 & 7-15-1+9 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{(Verified!)}$$

Step 3 🎯: Deduce A^{-1} and A^4

- Multiply by A^{-1}: $A^2 - 3A - I + 9A^{-1} = 0 \implies$

$$A^{-1} = -\frac{1}{9}(A^2 - 3A - I) = \frac{1}{9}\begin{bmatrix} 0 & 3 & 3 \\ 3 & 2 & -1 \\ 3 & -7 & -1 \end{bmatrix}$$

- Multiply by A: $A^4 = 3A^3 + A^2 - 9A$.

🎯 **Verified Result:** Cayley-Hamilton verified ($A^3 - 3A^2 - A + 9I = 0$) | $A^{-1} = -\frac{1}{9}(A^2 - 3A - I)$.

University Solved Examination Questions (Unit 1)

MMDU Official University Solved Examination Questions (Unit 1)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 **Question 1 (1 Mark): Rank of a Matrix & Row Echelon Form**

Question:
Define the rank of a matrix and state the rank of a matrix in Row Echelon Form.

Step-by-Step Solution:
The rank $\rho(A)$ of an $m \times n$ matrix A is the order of its largest non-vanishing minor (or the number of linearly independent rows). In Row Echelon Form, $\rho(A)$ equals the **total number of non-zero rows**.

🎯 **Verified Result:** $\rho(A)$ = Number of non-zero rows in echelon form.

📘 Question 2 (1 Mark): Eigenvalues of Upper Triangular Matrix

Question:

Find the sum and product of the eigenvalues of $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$.

Step-by-Step Solution:

Since A is upper triangular, its eigenvalues are its diagonal elements: $\lambda_1 = 3, \lambda_2 = 2, \lambda_3 = 5$.

- **Sum of eigenvalues** $= \text{Tr}(A) = 3 + 2 + 5 = 10$.
- **Product of eigenvalues** $= |A| = 3 \times 2 \times 5 = 30$.

🎯 **Verified Result:** Sum $= 10$, Product $= 30$.

📘 Question 13 (2 Marks): Inverse via Cayley-Hamilton Theorem

Question:

State Cayley-Hamilton Theorem. If $A^2 - 4A + 3I = 0$, find A^{-1} in terms of A and I.

Step-by-Step Solution:

Every square matrix satisfies its own characteristic equation $|A - \lambda I| = 0$.

Pre-multiplying $A^2 - 4A + 3I = 0$ by A^{-1}:

$$A - 4I + 3A^{-1} = 0 \implies A^{-1} = \frac{1}{3}(4I - A)$$

🎯 **Verified Result:** $A^{-1} = \frac{1}{3}(4I - A)$.

UNIT 2 • MMDU MULLANA ACADEMIC CURRICULUM

Unit 2: Unit II: Calculus (10 Hours)

Exam Weightage: 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Estimated Study Time:** 35 mins

Complete Unit II coverage per MMDU BMAT-001 syllabus: Rolle's Theorem, Mean value theorems, Taylor's and Maclaurin's series, Expansion by use of known series, Curvature, Radius of Curvature for Cartesian, Parametric and polar curves, Centre of Curvature, chord of Curvature. Indeterminate forms and L'Hospital's rule, Beta and Gamma Functions and their properties.

1. Mean Value Theorems & Power Series Expansions

1. In-Depth Analysis of the Three Fundamental Mean Value Theorems

Differential calculus is built on three interconnected foundation stones: Rolle's Theorem, Lagrange's Mean Value Theorem, and Cauchy's Generalized Mean Value Theorem.

A. Rolle's Theorem (1691)

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let $f : [a, b] \to \mathrm{R}$ be a real function satisfying:

1. f is **continuous** on the closed interval $[a, b]$.

2. f is **differentiable** on the open interval (a, b).

3. $f(a) = f(b)$.

> Then there exists at least one point $c \in (a, b)$ such that:

$$\mathbf{f'(c) = 0}$$

- **Geometric Interpretation:** If a continuous smooth curve starts and ends at the same horizontal height, there must be at least one crest or trough where the tangent line is perfectly horizontal (slope = 0).

- **Physical Interpretation:** If a car returns to its starting point after a road trip, its instantaneous velocity must have been zero at least once when turning around.

B. Lagrange's Mean Value Theorem (First MVT)

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let $f : [a, b] \to \mathrm{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists at least one point $c \in (a, b)$ such that:

$$\mathbf{f'(c) = \frac{f(b) - f(a)}{b - a}}$$

- **Geometric Interpretation:** There exists at least one interior point c where the instantaneous tangent line is strictly parallel to the chord (secant line) joining the endpoints $(a, f(a))$ and $(b, f(b))$.

- **Alternative Form:** Setting $b = a + h$ and $c = a + \theta h$ with $0 < \theta < 1$:

$$f(a + h) = f(a) + hf'(a + \theta h)$$

C. Cauchy's Generalized Mean Value Theorem

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let two functions $f, g : [a, b] \to \mathrm{R}$ be continuous on $[a, b]$, differentiable on (a, b), and $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists at least one $c \in (a, b)$ such that:

$$\mathbf{\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}}$$

- **Proof Architecture:** Define auxiliary function $\phi(x) = f(x) - \lambda g(x)$. Choose $\lambda = \frac{f(b)-f(a)}{g(b)-g(a)}$ so that $\phi(a) = \phi(b)$. Apply Rolle's theorem to $\phi(x)$ to obtain $\phi'(c) = f'(c) - \lambda g'(c) = 0$.

---

2. Taylor's Theorem with Remainder & Power Series

Taylor's Theorem with Remainder Terms:

If $f(x)$ and its first $(n-1)$ derivatives are continuous on $[a, a+h]$ and $f^{(n)}(x)$ exists on $(a, a+h)$, then:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n$$

Where the remainder R_n can be expressed as:

- **Lagrange's Form of Remainder:** $\mathbf{R_n = \frac{h^n}{n!}f^{(n)}(a+\theta h)}, \quad (0 < \theta < 1)$
- **Cauchy's Form of Remainder:** $\mathbf{R_n = \frac{h^n}{(n-1)!}(1-\theta)^{n-1}f^{(n)}(a+\theta h)}, \quad (0 < \theta < 1)$

Maclaurin's Series Expansion (Centering at $a = 0$):

When $\lim_{n\to\infty} R_n = 0$, the function can be expressed as an infinite power series:

$$\mathbf{f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \ldots}$$

Standard Expansion Catalog for Exams (Expansion by Use of Known Series):

1. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \ldots \quad (\forall x \in \mathrm{R})$
2. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \quad (\forall x \in \mathrm{R})$
3. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots \quad (\forall x \in \mathrm{R})$
4. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots \quad (-1 < x \leq 1)$
5. $(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \ldots \quad (|x| < 1)$

Problem:
Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on the interval $[1, 3]$.

Step 1 🚀: Check Rolle's Three Hypotheses

1. $f(x)$ is a polynomial, hence continuous on $[1, 3]$.

2. $f'(x) = 2x - 4$ exists on $(1, 3)$, hence differentiable on $(1, 3)$.

3. Check boundary values:

$$f(1) = 1^2 - 4(1) + 3 = 0$$

$$f(3) = 3^2 - 4(3) + 3 = 9 - 12 + 3 = 0 \implies f(1) = f(3)$$

Step 2 ⚡: Find Stationary Point $c \in (1, 3)$ where $f'(c) = 0$

$$f'(c) = 2c - 4 = 0 \implies \mathbf{c = 2}$$

Since $c = 2 \in (1, 3)$, Rolle's Theorem is **completely verified**!

🎯 Rolle's Theorem verified: $\mathbf{c = 2 \in (1, 3)}$.

University Question (4 Marks - MMDU Question Paper):
Expand $f(x) = e^x \cos x$ up to the term containing x^4 using Maclaurin's series (and verify by multiplication of known series).

Step 1 🚀: Compute Successive Derivatives at $x = 0$

- $f(x) = e^x \cos x \implies f(0) = 1$

- $f'(x) = e^x(\cos x - \sin x) \implies f'(0) = 1(1 - 0) = 1$

- $f''(x) = -2e^x \sin x \implies f''(0) = 0$

- $f'''(x) = -2e^x(\sin x + \cos x) \implies f'''(0) = -2(0 + 1) = -2$

- $f^{(4)}(x) = -4e^x \cos x \implies f^{(4)}(0) = -4(1) = -4$

Step 2 ⚡ : Substitute into Maclaurin's Formula (or Known Series)

$$e^x \cos x = (1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots)(1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots) = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \dots$$

🎯 Maclaurin Expansion: $\mathbf{e^x \cos x = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + O(x^5)}$.

2. Curvature, Centre & Chord of Curvature (Cartesian, Parametric, Polar)

1. Curvature, Radius & Center of Curvature: Comprehensive Theory

Physical Concept of Curvature (κ):

✅ **FUNDAMENTAL LAW / RULE:**
Definition: Let P and Q be two neighboring points on a smooth curve separated by arc length Δs. If the tangent vectors at P and Q make angles ψ and $\psi + \Delta\psi$ with the positive x-axis, the **Curvature** κ is the rate of turning of the tangent line per unit arc length:

$$\kappa = \lim_{\Delta s \to 0} \left| \frac{\Delta \psi}{\Delta s} \right| = \left| \frac{d\psi}{ds} \right|$$

Radius of Curvature (ρ):

The Radius of Curvature ρ is the reciprocal of the curvature:

$$\rho = \frac{1}{\kappa} = \left| \frac{ds}{d\psi} \right|$$

- A straight line has zero bending $\implies \kappa = 0 \implies \rho = \infty$.
- A circle of radius R bends uniformly $\implies \kappa = 1/R \implies \rho = R$.

---

2. Analytical Formulas for Radius of Curvature ρ

Coordinate Representation	Curve Equation	Radius of Curvature Formula ρ
Cartesian $y = f(x)$	$y = f(x)$	$\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$ $(y_1 = \frac{dy}{dx},\ y_2 = \frac{d^2y}{dx^2})$
Cartesian $x = g(y)$	$x = g(y)$	$\rho = \frac{(1+x_1^2)^{3/2}}{x_2}$ $(x_1 = \frac{dx}{dy},\ x_2 = \frac{d^2x}{dy^2})$
Parametric	$x = x(t),\ y = y(t)$	$\rho = \frac{(x'^2+y'^2)^{3/2}}{x'y''-y'x''}$
Polar Form	$r = f(\theta)$	$\rho = \frac{(r^2+r_1^2)^{3/2}}{r^2+2r_1^2-rr_2}$ $(r_1 = \frac{dr}{d\theta},\ r_2 = \frac{d^2r}{d\theta^2})$
Pedal Form (p-r)	$p = f(r)$	$\rho = r\frac{dr}{dp}$

---

3. Center of Curvature $(\bar{x}, \bar{y})$ and Chord of Curvature

The circle that shares the same tangent and curvature as the curve at point $P(x, y)$ is the **Osculating Circle (Circle of Curvature)**. Its center $C(\bar{x}, \bar{y})$ is given by:

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}, \quad \bar{y} = y + \frac{1+y_1^2}{y_2}$$

- **Equation of Circle of Curvature:** $(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$.
- **Evolute:** The locus of the centers of curvature of all points on a curve is called the **Evolute** of that curve.

Chords of Curvature:

- **Chord parallel to x-axis:** $C_x = 2\rho\sin\psi = \frac{2y_1(1+y_1^2)}{y_2}$
- **Chord parallel to y-axis:** $C_y = 2\rho\cos\psi = \frac{2(1+y_1^2)}{y_2}$
- **Chord through the Pole (Polar Curve $r = f(\theta)$):** $C_p = 2\rho\sin\phi = \frac{2r(r^2+r_1^2)}{r^2+2r_1^2-rr_2}$

University Question (6 Marks):
Find the radius of curvature at any point t on the cycloid $x = a(t+\sin t),\ y = a(1-\cos t)$.

Step 1 🚀: Compute First and Second Derivatives w.r.t t

- $x' = \frac{dx}{dt} = a(1+\cos t) = 2a\cos^2(\frac{t}{2})$

- $y' = \frac{dy}{dt} = a\sin t = 2a\sin(\frac{t}{2})\cos(\frac{t}{2})$

- $x'' = -a\sin t, \quad y'' = a\cos t$

Step 2 ⚡ : Denominator $x'y'' - y'x''$ and Numerator $(x'^2 + y'^2)^{3/2}$

$$x'y'' - y'x'' = a^2(1+\cos t) = 2a^2\cos^2(\frac{t}{2})$$

$$(x'^2 + y'^2)^{3/2} = [4a^2\cos^2(\frac{t}{2})]^{3/2} = 8a^3\cos^3(\frac{t}{2})$$

$$\rho = \frac{8a^3\cos^3(t/2)}{2a^2\cos^2(t/2)} = 4a\cos(\frac{t}{2})$$

🎯 Radius of Curvature: $\rho = \mathbf{4a}\cos(\frac{\mathbf{t}}{\mathbf{2}})$.

University Question (6 Marks - MMDU Question Paper):
Find the coordinates of the centre of curvature $(\bar{X}, \bar{Y})$ and the equation of the evolute of the parabola $y^2 = 4ax$.

Step 1 🚀 : Derivatives for $y^2 = 4ax$

$$2yy_1 = 4a \implies y_1 = \frac{2a}{y}, \quad y_2 = -\frac{4a^2}{y^3}$$

Step 2 ⚡ : Centre of Curvature Formula

$$\bar{X} = x - \frac{y_1(1+y_1^2)}{y_2} = x + \frac{y^2 + 4a^2}{2a} = 3x + 2a$$

$$\bar{Y} = y + \frac{1+y_1^2}{y_2} = -\frac{y^3}{4a^2}$$

Step 3 🎯: Eliminate (x, y) to Obtain the Evolute Equation
Substitute $x = \frac{\bar{X}-2a}{3}$ and $y^2 = (4a^2\bar{Y})^{2/3}$ into $y^2 = 4ax$:

$$27a\bar{Y}^2 = 4(\bar{X} - 2a)^3$$

🎯 Centre of Curvature: $(\bar{X}, \bar{Y}) = (3x + 2a, -\frac{y^3}{4a^2})$ | Evolute: $\mathbf{27aY^2 = 4(X - 2a)^3}$.

3. Indeterminate Forms & L'Hospital's Rule

1. Rigorous Analysis of Indeterminate Forms & L'Hospital's Rule

What is an Indeterminate Form?

An algebraic limit expression $\lim_{x\to a} \frac{f(x)}{g(x)}$ is called **indeterminate** if substituting the limiting point $x = a$ fails to yield a well-defined value because the limiting behaviors of the numerator and denominator conflict.

The 7 Classic Indeterminate Forms:

$$\frac{\mathbf{0}}{\mathbf{0}}, \quad \frac{\infty}{\infty}, \quad \mathbf{0 \times \infty}, \quad \infty - \infty, \quad \mathbf{0^0}, \quad \mathbf{1^\infty}, \quad \infty^{\mathbf{0}}$$

---

2. L'Hospital's Rule Theorem & Derivation

✅ **FUNDAMENTAL LAW / RULE:**
Theorem (Bernoulli & L'Hospital): Let f and g be differentiable on an open interval containing a (except possibly at a). If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$ (or both tend to $\pm\infty$), and $g'(x) \neq 0$ near a, then:

$$\lim_{\mathbf{x\to a}} \frac{\mathbf{f(x)}}{\mathbf{g(x)}} = \lim_{\mathbf{x\to a}} \frac{\mathbf{f'(x)}}{\mathbf{g'(x)}}$$

---

3. Systematic Reduction Strategies for Non-Fractional Forms

1. **Form $0 \times \infty$:** Invert one factor into the denominator: $f(x) \cdot g(x) = \frac{f(x)}{1/g(x)} \implies \frac{0}{0}$.

2. **Form $\infty - \infty$:** Combine over a common denominator using LCM or rationalization.

3. **Exponential Forms ($0^0, 1^\infty, \infty^0$):** Let $y = [f(x)]^{g(x)}$, take $\ln y = g(x)\ln[f(x)]$, evaluate $L = \lim_{x\to a} \ln y$, and recover $\lim y = e^L$.

University Question (4 Marks):
Evaluate $\lim_{x\to 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x}\right)$.

Step 1 🚀: Convert $\infty - \infty$ into $\frac{0}{0}$ via LCM

$$\lim_{x\to 0}\left(\frac{\sin^2 x - x^2}{x^2 \sin^2 x}\right) = \lim_{x\to 0} \frac{\sin^2 x - x^2}{x^4} \cdot \lim_{x\to 0}\left(\frac{x}{\sin x}\right)^2 = \lim_{x\to 0} \frac{\sin^2 x - x^2}{x^4}$$

Step 2 ⚡: Use Series Expansion $\sin x = x - \frac{x^3}{6} + O(x^5)$

$$\sin^2 x - x^2 = \left(x - \frac{x^3}{6}\right)^2 - x^2 = -\frac{x^4}{3} + O(x^6)$$

$$\lim_{x\to 0} \frac{-\frac{x^4}{3} + O(x^6)}{x^4} = -\frac{1}{3}$$

🎯 Evaluated Limit: $-\frac{1}{3}$.

University Question (6 Marks - MMDU Question Paper):
Evaluate $\lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{1/x^2}$.

Step 1 🚀: Identify 1^∞ Form & Take Natural Logarithm
Let $y = \lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{1/x^2} \implies \ln y = \lim_{x\to 0} \frac{\ln\left(\frac{\tan x}{x}\right)}{x^2}$.

Step 2 ⚡: Series Expansion of $\tan x$

$$\frac{\tan x}{x} = 1 + \frac{x^2}{3} + O(x^4) \implies \ln\left(\frac{\tan x}{x}\right) = \frac{x^2}{3} + O(x^4)$$

Step 3 🎯: Evaluate Limit and Exponentiate

$$\ln y = \lim_{x\to 0} \frac{\frac{x^2}{3} + O(x^4)}{x^2} = \frac{1}{3} \implies y = e^{1/3} = \sqrt[3]{e}$$

🎯 Limit: $\mathbf{e^{1/3} = \sqrt[3]{e}}$.

4. Beta and Gamma Functions & Their Properties

1. Rigorous Theory of Beta and Gamma Special Functions

A. The Gamma Function (Euler's Integral of Second Kind):

✅ **FUNDAMENTAL LAW / RULE:**
Definition: For any real number $n > 0$, the **Gamma Function** $\Gamma(n)$ is defined as:

$$\boldsymbol{\Gamma(n) = \int_0^\infty x^{n-1} e^{-x}\, dx \quad (n > 0)}$$

Fundamental Properties of $\Gamma(n)$:

1. **Recurrence Relation:** $\boldsymbol{\Gamma(n+1) = n\,\Gamma(n)}$ and $\boldsymbol{\Gamma(n+1) = n!}$ for positive integers.
2. **Base & Half-Integer Values:** $\Gamma(1) = 1$ and $\boldsymbol{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$.

---

2. The Beta Function (Euler's Integral of First Kind)

✅ **FUNDAMENTAL LAW / RULE:**
Definition: For $m > 0, n > 0$, the **Beta Function** $B(m, n)$ is defined as:

$$\boldsymbol{B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx}$$

Key Properties and Master Identities:

1. **Symmetry:** $\mathbf{B(m, n) = B(n, m)}$.

2. **Trigonometric Form:**

$$\mathbf{B(m, n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta}$$

3. **Infinite Domain Form:**

$$\mathbf{B(m, n) = \int_0^{\infty} \frac{t^{m-1}}{(1+t)^{m+n}} \, dt}$$

4. **Beta-Gamma Bridge Relation:**

$$\mathbf{B(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}}$$

5. **Walli's / Trigonometric Integral Formula:**

$$\int_0^{\pi/2} \sin^{p}\theta \cos^{q}\theta \, d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\,\Gamma\left(\frac{p+q+2}{2}\right)}$$

6. **Legendre's Duplication Formula:**

$$\mathbf{\Gamma(m)\,\Gamma\left(m+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}}\,\Gamma(2m)}$$

University Question (6 Marks - MMDU Question Paper):
Evaluate $\int_0^{\pi/2} \sin^4\theta \cos^6\theta \, d\theta$ using Beta and Gamma functions, and prove $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$.

Step 1 🚀: Evaluate Trigonometric Integral ($p = 4, q = 6$)

$$I_1 = \frac{\Gamma(\frac{5}{2})\Gamma(\frac{7}{2})}{2\,\Gamma(6)} = \frac{(\frac{3\sqrt{\pi}}{4})(\frac{15\sqrt{\pi}}{8})}{2 \times 120} = \frac{3\pi}{512}$$

Step 2 ⚡: Prove $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$
Substitute $x^4 = u \implies x = u^{1/4},\ dx = \frac{1}{4}u^{-3/4}du$:

$$I_2 = \frac{1}{4}\int_0^1 u^{1/4-1}(1-u)^{1/2-1}du = \frac{1}{4}B(\frac{1}{4},\frac{1}{2}) = \frac{\Gamma(1/4)\Gamma(1/2)}{4\,\Gamma(3/4)} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$$

🎯 Integral Value: $\frac{3\pi}{512}$ | Formula proved via $B(\frac{1}{4},\frac{1}{2})$.

University Solved Examination Questions (Unit 2)

MMDU Official University Solved Examination Questions (Unit 2)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 4 (1 Mark): Geometric Interpretation of Rolle's Theorem

Question:
State the geometric meaning of the point c in Rolle's Theorem.

Step-by-Step Solution:
If a continuous, differentiable curve $y = f(x)$ has equal values at the endpoints $f(a) = f(b)$, there exists at least one interior point $c \in (a, b)$ where $f'(c) = 0$, meaning the **tangent line at** $(c, f(c))$ **is parallel to the** x**-axis**.

🎯 **Verified Result:** Horizontal tangent parallel to x-axis at $x = c$.

📘 Question 14 (2 Marks): Radius of Curvature of Catenary Curve

Question:

Find the radius of curvature ρ for the catenary $y = c\cosh(x/c)$.

Step-by-Step Solution:

Given $y = c\cosh(x/c)$, we have $y_1 = \sinh(x/c)$ and $y_2 = \frac{1}{c}\cosh(x/c)$.
Since $1 + y_1^2 = 1 + \sinh^2(x/c) = \cosh^2(x/c)$:

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\cosh^3(x/c)}{\frac{1}{c}\cosh(x/c)} = c\cosh^2(x/c) = \frac{y^2}{c}$$

🎯 **Verified Result:** $\rho = \frac{y^2}{c}$.

📘 Question 22 (6 Marks): Relation Between Beta and Gamma Functions

Question:

State and prove $B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ and hence evaluate $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$.

Step-by-Step Solution:

Using polar coordinates $x = r\cos\theta, y = r\sin\theta$ in $\Gamma(m)\Gamma(n) = 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2m-1} y^{2n-1}\, dx\, dy$ yields $\Gamma(m)\Gamma(n) = B(m,n)\Gamma(m+n)$.
For $\int_0^{\pi/2} \sin^{1/2}\theta \cos^{-1/2}\theta\, d\theta$: here $p = 1/2, q = -1/2$, so:

$$I = \frac{\Gamma(3/4)\Gamma(1/4)}{2\Gamma(1)} = \frac{1}{2}\cdot\frac{\pi}{\sin(\pi/4)} = \frac{\pi}{\sqrt{2}}$$

🎯 **Verified Result:** $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta = \frac{\pi}{\sqrt{2}}$.

UNIT 3 • MMDU MULLANA ACADEMIC CURRICULUM

Unit 3: Unit III: Multivariable Calculus (Differentiation) (10 Hours)

Exam Weightage: 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Estimated Study Time:** 35 mins

Complete Unit III coverage per MMDU BMAT-001 syllabus: Limit, continuity and partial derivatives, Homogeneous Function, Euler's theorem, total derivative; Maxima and minima for the function of two variables; Method of Lagrange multipliers. Scalar and Vector functions, Gradient, Directional derivatives, divergence and curl.

1. Limits, Continuity & Partial Derivatives

1. Multivariable Functions: Limits, Continuity & Partial Derivatives

A. Limit of a Function of Two Variables:

✅ **FUNDAMENTAL LAW / RULE:**
Rigorous ϵ-δ Definition: We write $\lim_{(x,y)\to(a,b)} f(x,y) = L$ if for every $\epsilon > 0$, there exists a $\delta > 0$ such that:

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x,y) - L| < \epsilon$$

- **Path Independence Principle:** For a limit to exist in R^2, the function must approach the **exact same value** L **along EVERY conceivable path** approaching (a, b).
- **Two-Path Non-Existence Test:** If $f(x,y)$ yields different values along paths $y = mx$ or $y = mx^2$ (i.e., limit depends on m), the limit **strictly DOES NOT EXIST**!

B. Continuity of $f(x,y)$:

A function $f(x,y)$ is continuous at (a,b) if:

1. $f(a,b)$ is defined.
2. $\lim_{(x,y)\to(a,b)} f(x,y)$ exists.
3. $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$.

2. Partial Derivatives & Higher-Order Derivatives

$$\frac{\partial f}{\partial x} = f_x(x,y) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}, \quad \frac{\partial f}{\partial y} = f_y(x,y) = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}$$

✅ **FUNDAMENTAL LAW / RULE:**

Clairaut's Symmetry Theorem: If the mixed partial derivatives f_{xy} and f_{yx} are both **continuous** on an open region containing (a, b), then:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} \iff f_{yx} = f_{xy}$$

University Question (4 Marks):

Evaluate whether $\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4+y^2}$ exists.

Step 1 🚀: Approach along straight line paths $y = mx$

$$\lim_{x\to 0} \frac{x^2(mx)}{x^4 + m^2x^2} = \lim_{x\to 0} \frac{mx}{x^2 + m^2} = 0$$

Step 2 ⚡: Approach along parabolic path $y = mx^2$

$$\lim_{x\to 0} \frac{x^2(mx^2)}{x^4 + m^2x^4} = \frac{m}{1+m^2}$$

Since the limit depends on m, the limit is **not unique** and **does NOT exist**.

🎯 By Path Test ($y = mx^2$), the limit **does NOT exist**.

University Question (6 Marks - MMDU Question Paper):

If $u = \ln(x^2 + y^2)$, prove that u satisfies 2D Laplace's Equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Step 1 🚀: First Partial Derivatives

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2+y^2}, \quad \frac{\partial u}{\partial y} = \frac{2y}{x^2+y^2}$$

Step 2 ⚡: Second Partial Derivatives via Quotient Rule

$$\frac{\partial^2 u}{\partial x^2} = \frac{2(x^2+y^2)-2x(2x)}{(x^2+y^2)^2} = \frac{2y^2-2x^2}{(x^2+y^2)^2}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{2x^2-2y^2}{(x^2+y^2)^2}$$

Step 3 🎯: Sum the Second Derivatives

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{2y^2-2x^2+2x^2-2y^2}{(x^2+y^2)^2} = 0$$

🎯 Proved: $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ (u is Harmonic).

2. Euler's Theorem on Homogeneous Functions

1. Homogeneous Functions & Euler's Theorem: Rigorous Analysis

✅ **FUNDAMENTAL LAW / RULE:**
Definition: A multivariable function $f(x,y)$ is **homogeneous of degree** n if for any scalar $t > 0$:

$$\mathbf{f(tx, ty) = t^n f(x, y) \iff f(x, y) = x^n \phi\left(\frac{y}{x}\right)}$$

2. Euler's Theorem on Homogeneous Functions (First & Second Order)

✅ **FUNDAMENTAL LAW / RULE:**
First-Order Euler's Theorem: If $u = f(x,y)$ is a homogeneous function of x and y of degree n with continuous partial derivatives, then:

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=n\,u$$

✅ **FUNDAMENTAL LAW / RULE:**
Second-Order Euler's Formula:

$$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=n(n-1)u$$

🚨 **EXAMINER TRAP / PITFALL:**
Examiner's Composite Function Deduction: If $z=f(u)$ is homogeneous of degree n in (x,y), then:

1. $xu_x+yu_y=n\frac{f(u)}{f'(u)}=g(u)$

2. $x^2u_{xx}+2xyu_{xy}+y^2u_{yy}=g(u)\,[g'(u)-1]$

University Question (6 Marks - MMDU Question Paper):

If $u=\sin^{-1}\left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}\right)$, prove that $xu_x+yu_y+zu_z=-3\tan u$.

Step 1 🚀: Convert to Homogeneous Function $v=\sin u$

$$v=\sin u=\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$$

Degree of numerator is 1; degree of denominator is $\sqrt{t^8}=t^4$. Thus v is homogeneous of degree $n=1-4=-3$.

Step 2 ⚡: Apply 3-Variable Euler's Theorem to v

$$x\frac{\partial v}{\partial x}+y\frac{\partial v}{\partial y}+z\frac{\partial v}{\partial z}=-3v$$

Using $\frac{\partial v}{\partial x} = \cos u \frac{\partial u}{\partial x}$:

$$\cos u\,(xu_x + yu_y + zu_z) = -3\sin u \implies xu_x + yu_y + zu_z = -3\tan u$$

🎯 Result Proved: $\mathbf{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = -3\tan u}$.

University Question (6 Marks - MMDU Previous Paper):
If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u\,(2\cos 2u - 1) = \sin 4u - \sin 2u$.

Step 1 🚀: Define $z = \tan u$ (Degree $n = 3 - 1 = 2$)

$$xz_x + yz_y = 2z \implies x\sec^2 u\,u_x + y\sec^2 u\,u_y = 2\tan u \implies xu_x + yu_y = \frac{2\tan u}{\sec^2 u} = \sin 2u = g(u)$$

Step 2 ⚡: Second-Order Euler Identity

$$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1] = \sin 2u\,(2\cos 2u - 1) = \sin 4u - \sin 2u$$

🎯 Second-Order Euler Identity Proved: $\mathbf{x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(2\cos 2u - 1)}$.

3. Total Derivative, Maxima-Minima & Lagrange Multipliers

1. Total Differential, Chain Rule & Extrema of Multivariable Functions

A. The Total Differential (du) & Total Derivative ($\frac{du}{dt}$):

- **Total Differential:** $du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz$
- **Total Derivative (Single Parameter t):**

$$\frac{\mathbf{du}}{\mathbf{dt}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\frac{\mathbf{dx}}{\mathbf{dt}} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\frac{\mathbf{dy}}{\mathbf{dt}}$$

- **Composite Chain Rule ($x = x(r,\theta),\ y = y(r,\theta)$):**

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}, \quad \frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta}$$

---

2. Unconstrained Maxima & Minima of Two Variables ($rt - s^2$ Test)

To locate the local maxima, local minima, and saddle points of $z = f(x, y)$:

1. **Stationary Points:** Solve $f_x = 0$ and $f_y = 0$.

2. **Second-Order Partials at (a, b):** $r = f_{xx},\ s = f_{xy},\ t = f_{yy}$ and Discriminant $\Delta = rt - s^2$:

Condition on $\Delta = rt - s^2$	Condition on $r = f_{xx}$	Nature of Critical Point (a, b)
$rt - s^2 > 0$	$r < 0$	**Local Maximum**
$rt - s^2 > 0$	$r > 0$	**Local Minimum**
$rt - s^2 < 0$	Any sign	**Saddle Point**
$rt - s^2 = 0$	Indeterminate	**Test Fails** (Further investigation required)

---

3. Constrained Optimization: Method of Lagrange Multipliers

To optimize $f(x, y, z)$ subject to constraint $\phi(x, y, z) = 0$:

1. Form Auxiliary Lagrangian: $F(x, y, z, \lambda) = f(x, y, z) + \lambda\, \phi(x, y, z)$.

2. Solve the stationary system:

$$f_x + \lambda \phi_x = 0, \quad f_y + \lambda \phi_y = 0, \quad f_z + \lambda \phi_z = 0, \quad \phi(x, y, z) = 0$$

University Question (4 Marks):
Find the local maximum and minimum values of $f(x, y) = x^3 + y^3 - 3xy$.

Step 1 🚀: Find Critical (Stationary) Points
$f_x = 3x^2 - 3y = 0 \implies y = x^2$ and $f_y = 3y^2 - 3x = 0 \implies 3x(x^3 - 1) = 0$.
Critical points: $(0, 0)$ and $(1, 1)$.

Step 2 ⚡: Compute Second Partials r, s, t

$r = f_{xx} = 6x, \quad s = f_{xy} = -3, \quad t = f_{yy} = 6y \implies \Delta = rt - s^2 = 36xy - 9.$

Step 3 🎯: Classify Points

1. **At** $(0,0)$**:** $\Delta = 0 - 9 = -9 < 0 \implies (0,0)$ is a **Saddle Point**.

2. **At** $(1,1)$**:** $\Delta = 36 - 9 = 27 > 0$ and $r = 6 > 0 \implies (1,1)$ is a **Local Minimum** with $f_{\min}(1,1) = 1 + 1 - 3 = -1$.

🎯 Saddle point: $(0,0)$ | Local Minimum at $(1,1)$ with $\mathbf{f_{min} = -1}$.

University Question (6 Marks - MMDU Question Paper):
Find the maximum volume of a rectangular box open at the top, having a given surface area $S = 108 \text{cm}^2$, using Lagrange's method of multipliers.

Step 1 🚀: Objective and Constraint Functions
Maximize $V = xyz$ subject to $\phi(x,y,z) = xy + 2yz + 2zx - 108 = 0$.
Lagrangian: $F(x,y,z,\lambda) = xyz + \lambda(xy + 2yz + 2zx - 108)$.

Step 2 ⚡: Set First Partials to Zero

$$yz + \lambda(y + 2z) = 0, \quad xz + \lambda(x + 2z) = 0, \quad xy + \lambda(2x + 2y) = 0$$

Equating λ gives $x = y = 2z \implies z = \frac{x}{2}$.

Step 3 🎯: Substitute into Constraint $S = 108$

$$x^2 + 2x(x/2) + 2x(x/2) = 3x^2 = 108 \implies x = 6,\ y = 6,\ z = 3 \implies V_{\max} = 6 \times 6 \times 3 = 108 \text{cm}^3$$

🎯 Dimensions: $6\text{cm} \times 6\text{cm} \times 3\text{cm}$ | Maximum Volume: $\mathbf{V_{max} = 108 \text{cm}^3}$.

4. Scalar & Vector Functions, Gradient, Divergence & Curl

1. Vector Differential Calculus: Del, Gradient, Divergence & Curl

✅ **FUNDAMENTAL LAW / RULE:**
The Vector Differential Operator Del (∇):

$$\nabla = \hat{\mathbf{i}}\frac{\partial}{\partial \mathbf{x}} + \hat{\mathbf{j}}\frac{\partial}{\partial \mathbf{y}} + \hat{\mathbf{k}}\frac{\partial}{\partial \mathbf{z}}$$

A. Gradient of a Scalar Field ($\nabla\phi$) & Directional Derivative:

- **Gradient:** $\nabla\phi = \hat{\mathbf{i}}\frac{\partial\phi}{\partial \mathbf{x}} + \hat{\mathbf{j}}\frac{\partial\phi}{\partial \mathbf{y}} + \hat{\mathbf{k}}\frac{\partial\phi}{\partial \mathbf{z}}$ (normal to level surface $\phi(x, y, z) = c$).
- **Unit Normal Vector:** $\hat{\mathbf{n}} = \frac{\nabla\phi}{|\nabla\phi|}$.
- **Directional Derivative along $\vec{a}$:** $\text{D.D.} = \nabla\phi \cdot \frac{\vec{\mathbf{a}}}{|\vec{\mathbf{a}}|}$ (maximum magnitude = $|\nabla\phi|$ in direction of $\nabla\phi$).

B. Divergence ($\nabla \cdot \vec{F}$) & Curl ($\nabla \times \vec{F}$):

- **Divergence (Scalar):** $\text{div}\,\vec{\mathbf{F}} = \nabla \cdot \vec{\mathbf{F}} = \frac{\partial \mathbf{F_1}}{\partial \mathbf{x}} + \frac{\partial \mathbf{F_2}}{\partial \mathbf{y}} + \frac{\partial \mathbf{F_3}}{\partial \mathbf{z}}$.

- **Solenoidal Field:** $\nabla \cdot \vec{\mathbf{F}} = \mathbf{0}$.

- **Curl (Vector):** $\text{curl}\,\vec{\mathbf{F}} = \nabla \times \vec{\mathbf{F}} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$.

- **Irrotational (Conservative) Field:** $\nabla \times \vec{\mathbf{F}} = \vec{\mathbf{0}} \iff \vec{\mathbf{F}} = \nabla\phi$.

University Question (6 Marks - MMDU Question Paper):
Find the directional derivative of $\phi = x^2yz + 4xz^2$ at point $P(1, -2, -1)$ in the direction of vector $\vec{a} = 2\hat{i} - \hat{j} - 2\hat{k}$.

Step 1 🚀: Calculate $\nabla\phi$ at $P(1, -2, -1)$

$$\nabla\phi = (2xyz + 4z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 8xz)\hat{k}\Big|_{(1,-2,-1)} = 8\hat{i} - \hat{j} - 10\hat{k}$$

Step 2 ⚡: Project onto Unit Vector $\hat{a}$

$$\hat{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{4 + 1 + 4}} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3} \implies \text{D.D.} = \nabla\phi \cdot \hat{a} = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

🎯 Directional Derivative: $\frac{37}{3} \approx \mathbf{12.33}$.

University Question (6 Marks - MMDU Question Paper):
Prove that $\vec{F} = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$ is irrotational, and find its scalar potential ϕ.

Step 1 🚀: Evaluate $\text{curl}\, \vec{F} = \nabla \times \vec{F}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 \cos x + z^3 & 2y \sin x - 4 & 3xz^2 + 2 \end{vmatrix} = (0)\hat{i} - (3z^2 - 3z^2)\hat{j} + (2y \cos x - 2y \cos x)\hat{k} = \vec{0}$$

Step 2 ⚡: Integrate Exact Differential $d\phi = \vec{F} \cdot d\vec{r}$

$$d\phi = d(y^2 \sin x) + d(xz^3) - d(4y) + d(2z) \implies \phi = y^2 \sin x + xz^3 - 4y + 2z + C$$

🎯 $\nabla \times \vec{F} = \vec{0}$ (Irrotational) | Scalar Potential: $\phi = \mathbf{y^2 \sin x + xz^3 - 4y + 2z + C}$.

University Solved Examination Questions (Unit 3)

MMDU Official University Solved Examination Questions (Unit 3)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 7 (1 Mark): Homogeneous Degree via Euler's Theorem

Question:
Find the degree of homogeneity for $u(x,y) = \frac{x^3+y^3}{x-y}$ and evaluate $xu_x + yu_y$.

Step-by-Step Solution:
Since $u(tx, ty) = \frac{t^3(x^3+y^3)}{t(x-y)} = t^2 u(x,y)$, the degree is $n = 2$. By Euler's theorem, $xu_x + yu_y = 2u$.

🎯 **Verified Result:** Degree $n = 2$ | $xu_x + yu_y = 2u$.

📘 Question 19 (4 Marks): Two-Variable Extrema Determination

Question:

Find and classify all local extrema and saddle points of $f(x,y) = x^3 + y^3 - 3axy$ $(a > 0)$.

Step-by-Step Solution:

Stationary system: $f_x = 3x^2 - 3ay = 0$ and $f_y = 3y^2 - 3ax = 0 \implies$ critical points $(0,0)$ and (a,a).
With $r = 6x, s = -3a, t = 6y$ and $\Delta = rt - s^2 = 36xy - 9a^2$:

- At $(0,0)$: $\Delta = -9a^2 < 0 \implies$ **Saddle Point**.

- At (a,a): $\Delta = 27a^2 > 0, r = 6a > 0 \implies$ **Local Minimum** with $f_{\min} = -a^3$.

🎯 **Verified Result:** $(0,0)$ is a Saddle Point; (a,a) is a Local Minimum with value $-a^3$.

📘 Question 23 (6 Marks): Constrained Optimization via Lagrange Multipliers

Question:

Find the maximum and minimum distances from the origin to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ $(a > b > c > 0)$.

Step-by-Step Solution:

Using $F = (x^2 + y^2 + z^2) + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right)$, setting $F_x = F_y = F_z = 0$ yields $\lambda = -a^2, -b^2, -c^2$, corresponding to distances $d_{\max} = a$ and $d_{\min} = c$.

🎯 **Verified Result:** Maximum distance $= a$ | Minimum distance $= c$.

UNIT 4 • MMDU MULLANA ACADEMIC CURRICULUM

Unit 4: Unit IV: Sequences & Series (05 Hrs) and Fourier Series (05 Hrs)

Exam Weightage: 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Estimated Study Time:** 40 mins

Complete Unit IV coverage per MMDU BMAT-001 syllabus: Sequences and series (05 hours): Convergence of sequence and series; Tests for convergence: Comparison test, Ratio Test, Raabe's Test, Logarithmic Test, Cauchy root test, Alternating Series; Leibnitz's rule, conditionally convergent, absolute convergence. Fourier series (05 hours): Euler's formulae, Conditions for Fourier expansions, Fourier expansions of functions having point of discontinuity, change of interval, Odd & Even functions, Half-range series.

1. Convergence Tests: Comparison, Ratio, Raabe's & Logarithmic

1. In-Depth Theory of Infinite Series Convergence & Divergence

Let $\sum_{n=1}^{\infty} u_n = u_1 + u_2 + \cdots + u_n + \ldots$ be an infinite series with partial sums $S_k = \sum_{n=1}^{k} u_n$.

- **Necessary Condition (n^{th} Term Test):** If $\sum u_n$ converges, then $\lim_{\mathbf{n}\to\infty} \mathbf{u_n} = \mathbf{0}$. If $\lim_{n\to\infty} u_n \neq 0$, the series **strictly diverges**.

- **Geometric Series $\sum r^n$:** Converges iff $|r| < 1$.

- **Harmonic p-Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$:** Converges strictly if $\mathbf{p > 1}$, and diverges if $\mathbf{p \leq 1}$.

---

2. Systematic Positive-Term Convergence Tests Pipeline

1. **Limit Comparison Test:** If $\lim_{n\to\infty} \frac{u_n}{v_n} = l$ $(0 < l < \infty$, finite and non-zero), then $\sum u_n$ and $\sum v_n$ converge or diverge together.

2. **D'Alembert's Ratio Test:** Let $\mathbf{L} = \lim_{\mathbf{n}\to\infty} \frac{\mathbf{u_n}}{\mathbf{u_{n+1}}}$:

- $\mathbf{L > 1} \implies$ **Convergent** | $\mathbf{L < 1} \implies$ **Divergent** | $\mathbf{L = 1} \implies$ **Test Fails** (use Raabe's or Logarithmic test).

3. **Raabe's Test (when $L = 1$):** Evaluate $\mathbf{R} = \lim_{\mathbf{n}\to\infty} \mathbf{n}\left(\frac{\mathbf{u_n}}{\mathbf{u_{n+1}}} - \mathbf{1}\right)$:

- $\mathbf{R > 1} \implies$ **Convergent** | $\mathbf{R < 1} \implies$ **Divergent** | $\mathbf{R = 1} \implies$ **Test Fails**.

4. **Logarithmic Test (when $L = 1$ with exponents/e):** Evaluate $\mathbf{L'} = \lim_{\mathbf{n}\to\infty} \mathbf{n} \ln\left(\frac{\mathbf{u_n}}{\mathbf{u_{n+1}}}\right)$:

- $\mathbf{L' > 1} \implies$ **Convergent** | $\mathbf{L' < 1} \implies$ **Divergent**.

University Question (6 Marks - Classic MMDU Exam Problem):
Examine the convergence of the series:

$$\frac{1}{2}+\frac{1\cdot 3}{2\cdot 4}+\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}+\cdots+\frac{1\cdot 3\cdot 5\ldots(2n-1)}{2\cdot 4\cdot 6\ldots(2n)}+\ldots$$

Step 1 🚀: Ratio Test Setup

Here $\frac{u_n}{u_{n+1}}=\frac{2n+2}{2n+1} \implies L=\lim_{n\to\infty}\frac{2n+2}{2n+1}=1 \implies$ **Ratio Test Fails!**

Step 2 🗲: Apply Raabe's Test

$$R=\lim_{n\to\infty} n\left(\frac{2n+2}{2n+1}-1\right)=\lim_{n\to\infty}\frac{n}{2n+1}=\frac{1}{2}<1$$

🎯 Ratio test fails ($L=1$); Raabe's Test yields $R=\frac{1}{2}<1 \implies$ Series is **Divergent**.

2. Cauchy's Root Test, Alternating Series & Leibnitz's Rule

1. Cauchy's Root Test, Alternating Series & Leibnitz's Rule

A. Cauchy's n^{th} Root Test:

✅ **FUNDAMENTAL LAW / RULE:**

Theorem: Let $\sum u_n$ be a positive-term series and evaluate $\mathbf{L}=\lim_{\mathbf{n}\to\infty}(\mathbf{u_n})^{\mathbf{1/n}}$:

- $\mathbf{L<1} \implies$ **Convergent** | $\mathbf{L>1} \implies$ **Divergent** | $\mathbf{L=1} \implies$ **Test Fails**.

B. Alternating Series & Leibnitz's Rule:

✅ **FUNDAMENTAL LAW / RULE:**

Theorem: An alternating series $\sum_{n=1}^{\infty}(-1)^{n-1}u_n=u_1-u_2+u_3-u_4+\ldots$ $(u_n>0)$ converges if:

1. **Monotonic Decrease:** $u_{n+1}\le u_n$ for all n.

2. **Vanishing Limit:** $\lim_{n\to\infty}u_n=0$.

C. Absolute vs Conditional Convergence:

- **Absolutely Convergent:** $\sum |u_n|$ converges (e.g., $\sum \frac{(-1)^{n-1}}{n^2}$).
- **Conditionally Convergent:** $\sum u_n$ converges via Leibnitz's test, but $\sum |u_n|$ diverges (e.g., $\sum \frac{(-1)^{n-1}}{n}$ and $\sum \frac{(-1)^{n-1}}{\sqrt{n}}$).

University Question (4 Marks - MMDU Question Paper):
Test the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ for absolute and conditional convergence.

Step 1 🚀: Verify Leibnitz's Conditions
Here $u_n = \frac{1}{\sqrt{n}} > 0$. Since $\sqrt{n+1} > \sqrt{n} \implies u_{n+1} < u_n$, and $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$, the series **converges** by Leibnitz's Rule.

Step 2 ⚡: Check Absolute Series $\sum |u_n|$

$$\sum_{n=1}^{\infty} |u_n| = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$$

Since $p = \frac{1}{2} \leq 1$, $\sum |u_n|$ **diverges**. Therefore, the series is **Conditionally Convergent**.

🎯 $\sum \frac{(-1)^{n-1}}{\sqrt{n}}$ is **Conditionally Convergent** | $\sum \frac{(-1)^{n-1}}{n^2}$ is **Absolutely Convergent**.

3. Fourier Series: Euler's Formulae & Discontinuity

1. Fourier Series: Foundations & Euler's Formulas

Fourier Representation in $(-\pi, \pi)$ or $(0, 2\pi)$:

$$\mathbf{f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]}$$

$$\mathbf{a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\,dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\sin(nx)\,dx}$$

Dirichlet's Conditions & Jump Discontinuity:

At any finite jump discontinuity $x = c$, the Fourier series converges to the average of the left and right limits:

$$\mathbf{S(c) = \frac{f(c^+) + f(c^-)}{2}}$$

University Question (6 Marks - MMDU Question Paper):
Obtain the Fourier series for $f(x) = x^2$ in $(-\pi, \pi)$. Hence deduce $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Step 1 🚀: Symmetry & Coefficients
Since $f(-x) = (-x)^2 = x^2 = f(x)$, $f(x)$ is **even** $\implies b_n = 0$.

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x^2\,dx = \frac{2\pi^2}{3}, \quad a_n = \frac{2}{\pi}\int_0^{\pi} x^2\cos(nx)\,dx = \frac{4(-1)^n}{n^2}$$

Step 2 ⚡: Fourier Expansion & Deduction at $x = \pi$

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(nx) \implies \text{At } x = \pi: \ \pi^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{1}{n^2} \implies \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}$$

🎯 Fourier Series: $x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(nx)$ | Deduced: $\sum_{\mathbf{n=1}}^{\infty}\frac{\mathbf{1}}{\mathbf{n^2}} = \frac{\pi^2}{6}$.

4. Even/Odd Symmetries, Change of Interval & Half-Range Series

1. Even/Odd Symmetries, Change of Interval $(-l, l)$ & Half-Range Series

Parity / Domain	Fourier Coefficients	Series Expansion
Even Function in $(-l, l)$	$b_n = 0,\ a_0 = \frac{2}{l}\int_0^l f(x)dx,\ a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$
Odd Function in $(-l, l)$	$a_0 = 0, a_n = 0,\ b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$	$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$
Half-Range Cosine in $(0, l)$	$b_n = 0,\ a_0 = \frac{2}{l}\int_0^l f(x)dx,\ a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$	Even periodic extension of period $2l$
Half-Range Sine in $(0, l)$	$a_0 = 0, a_n = 0,\ b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$	Odd periodic extension of period $2l$

University Question (6 Marks - MMDU Question Paper):
Find the Fourier series for the triangular wave $f(x) = l - |x|$ in the arbitrary interval $(-l, l)$. Hence deduce $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$.

Step 1 🚀 : Symmetry Check in $(-l, l)$
Since $f(-x) = l - |-x| = f(x)$, $f(x)$ is **even** $\implies$ $b_n = 0$ and $a_0 = \frac{2}{l}\int_0^l (l-x)dx = l$.

Step 2 ⚡ : Compute a_n **and Deduce at** $x = 0$

$$a_n = \frac{2}{l}\int_0^l (l-x)\cos\left(\frac{n\pi x}{l}\right)dx = \frac{2l}{n^2\pi^2}[1-(-1)^n] = \begin{cases} \frac{4l}{n^2\pi^2}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

At $x = 0$: $l = \frac{l}{2} + \frac{4l}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right) \implies \sum_{n=1,3,5\dots}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{8}$.

🎯 Fourier Series: $\mathbf{l - |x| = \frac{l}{2} + \frac{4l}{\pi^2}\sum_{n=1,3,5\dots}\frac{1}{n^2}\cos\left(\frac{n\pi x}{l}\right)}$ | Deduced: $\mathbf{\sum_{n=1,3,5\dots}\frac{1}{n^2} = \frac{\pi^2}{8}}$.

University Solved Examination Questions (Unit 4)

MMDU Official University Solved Examination Questions (Unit 4)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 **Question 10 (1 Mark): Failure of Ratio Test on p-Series**

Question:
Why does D'Alembert's Ratio Test fail on the p-series $\sum \frac{1}{n^p}$?

Step-by-Step Solution:
Because $L = \lim_{n\to\infty}\frac{u_n}{u_{n+1}} = \lim_{n\to\infty}(1+\frac{1}{n})^p = 1^p = 1$ for every p, so the Ratio Test provides no conclusion and Raabe's Test must be used.

🎯 **Verified Result:** $L = 1$ for all p, hence Ratio Test fails.

📘 Question 20 (4 Marks): Raabe's Test on Higher-Order Series

Question:

Test the convergence of $1 + \frac{1}{2} \cdot \frac{1}{3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{1}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{1}{7} + \ldots$

Step-by-Step Solution:

Here $\frac{u_n}{u_{n+1}} = \frac{2n+2}{2n+1} \cdot \frac{2n+3}{2n+1} = \frac{4n^2+10n+6}{4n^2+4n+1} \to 1$ (Ratio Test fails).

Applying Raabe's Test:

$$R = \lim_{n \to \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \lim_{n \to \infty} \frac{6n^2 + 5n}{4n^2 + 4n + 1} = \frac{6}{4} = \frac{3}{2} > 1$$

Hence the series is **Convergent**.

🎯 **Verified Result:** Convergent by Raabe's Test ($R = 1.5 > 1$).

📘 Question 24 (6 Marks): Full Fourier Expansion of Discontinuous Square Wave

Question:

Obtain the Fourier series expansion for $f(x) = -k$ on $(-\pi, 0)$ and $f(x) = +k$ on $(0, \pi)$. Hence deduce $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$.

Step-by-Step Solution:

Since $f(x)$ is odd, $a_0 = 0, a_n = 0$, and $b_n = \frac{2}{\pi} \int_0^{\pi} k \sin(nx) dx = \frac{2k}{n\pi}[1 - (-1)^n] = \frac{4k}{n\pi}$ for odd n.

$$f(x) = \frac{4k}{\pi} \left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \ldots \right)$$

Substituting $x = \frac{\pi}{2}$ where $f(\pi/2) = k$ yields $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$.

🎯 **Verified Result:** $f(x) = \frac{4k}{\pi} \sum_{n=1,3,5\ldots} \frac{\sin nx}{n}$ | Gregory-Leibniz series $\frac{\pi}{4}$ proved.